

MODERN NETWORKING

Practical file

MSc Part 1

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Configure IP SLA Tracking and Path Control Topology

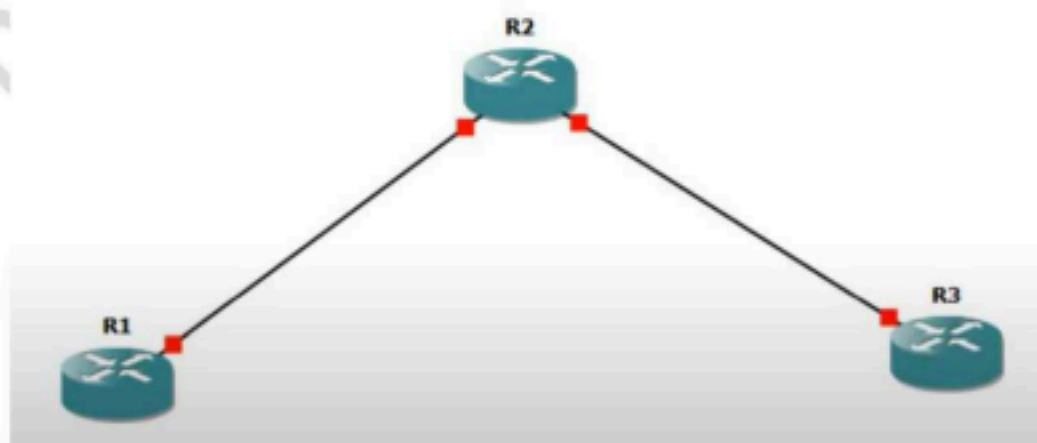
Aim: To configure and verify IP SLA (Service Level Agreement) tracking and path control on Cisco routers, enabling dynamic failover between multiple ISPs based on real-time network performance metrics.

Theory: IP SLA is a Cisco IOS feature that allows active monitoring of network performance by generating traffic and measuring parameters like delay, jitter, and packet loss. This proactive monitoring helps in assessing the quality of network paths.

Path Control involves directing traffic over specific network paths based on performance metrics. By integrating IP SLA with tracking objects, routers can make intelligent routing decisions, ensuring optimal path selection and network resilience.

Tracking Objects are used to monitor the status of IP SLA operations. If an IP SLA operation fails (e.g., due to high latency or packet loss), the tracking object reflects this state, allowing the router to adjust its routing decisions accordingly.

Topology:



Commands:**Configure IP SLA Operations**

```
R1(config)# ip sla 1
```

```
R1(config-ip-sla)# icmp-echo 209.165.200.254 source-interface Serial0/0/0
```

```
R1(config-ip-sla-echo)# timeout 5000
```

```
R1(config-ip-sla-echo)# frequency 10
```

```
R1(config)# ip sla schedule 1 life forever start-time now
```

Explanation: This configuration sets up an ICMP echo operation (ping) to the IP address 209.165.200.254, using Serial0/0/0 as the source interface. The operation has a timeout of 5000 ms and a frequency of 10 seconds.

Configure Tracking Objects

```
R1(config)# track 1 ip sla 1 reachability
```

```
R1(config-track)# delay down 10 up 1
```

```
R1(config-track)# exit
```

Explanation: This command creates a tracking object (ID 1) that monitors the reachability of IP SLA operation 1. The delay parameters ensure that transient failures don't cause unnecessary route changes.

Configure Static Routes with Tracking

```
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.201.1 5 track 1
```

```
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.202.129 10 track 2
```

Explanation: These commands set up two default routes with different administrative distances. The first route uses 209.165.201.1 with a distance of 5 and is tracked by object 1. The second route uses 209.165.202.129 with a distance of 10 and is tracked by object 2.

Configure IP SLA Responder on Remote Router

```
R2(config)# ip sla responder
```

Explanation: This command enables the IP SLA responder on the remote router (R2), allowing it to respond to the ICMP echo requests sent by R1.

Verification:

To verify the IP SLA operation:

```
R1# show ip sla statistics
```

To check the status of tracking objects:

```
R1# show track
```

To view the routing table and confirm active routes:

```
R1# show ip route
```

Conclusion:

By configuring IP SLA tracking and path control, the network can dynamically adjust to changing conditions, ensuring optimal path selection and improved network resilience. This setup is particularly beneficial in scenarios where multiple ISPs are used, allowing for automatic failover in case of link degradation or failure.

For video demonstration of the above practical scan the following QR-code or type the link address

<https://youtu.be/mlj40NPOH9I?si=0XWBVaZ27wcaacYt>



Using the AS_PATH Attribute

Aim: To understand and demonstrate the use of the AS_PATH attribute in Border Gateway Protocol (BGP) for path selection and traffic engineering by simulating a multi-AS network topology using GNS3.

Theory: The Border Gateway Protocol (BGP) is a standardized exterior gateway protocol designed to exchange routing and reachability information among autonomous systems (ASes) on the Internet. BGP is classified as a path-vector routing protocol, and it makes routing decisions based on paths, network policies, or rule-sets configured by a network administrator.

One of the key attributes in BGP is the AS_PATH attribute. This attribute lists the sequence of ASes that routing information has traversed. It serves two primary purposes:

Loop Prevention: By examining the AS_PATH, a BGP router can detect routing loops. If a router sees its own AS number in the AS_PATH of a received route, it will reject that route to prevent a loop.

Path Selection: BGP prefers routes with shorter AS_PATHs, assuming all other attributes are equal. This preference helps in selecting the most efficient path to a destination.

Additionally, network administrators can manipulate the AS_PATH attribute using a technique called AS Path Prepending. This involves adding multiple instances of an AS number to the AS_PATH to make a route less attractive, thereby influencing the path selection process.

Topology:



Code: sa

In the GNS3 simulation, we set up a network topology with three routers, each representing a different AS:

Router R1: AS 100

Router R2: AS 200

Router R3: AS 300

The goal is to advertise a network from R1 and observe how AS_PATH influences the route selection on R3.

Step 1: Configure BGP on R1

```
router bgp 100
network 192.168.1.0 mask 255.255.255.0
neighbor 10.0.12.2 remote-as 200
```

Step 2: Configure BGP on R2

```
router bgp 200
neighbor 10.0.12.1 remote-as 100
neighbor 10.0.23.3 remote-as 300
```

Step 3: Configure BGP on R3

```
router bgp 300
neighbor 10.0.23.2 remote-as 200
```

Step 4: Verify AS_PATH on R3

On R3, use the following command to view the BGP table:

```
show ip bgp
```

We observe the route to 192.168.1.0/24 with an AS_PATH of 200 100, indicating that the route has traversed AS 200 and AS 100.

Step 5: Apply AS Path Prepending on R2

To make the path through R2 less preferred, prepend AS 200 multiple times:

```
router bgp 200
neighbor 10.0.23.3 remote-as 300
neighbor 10.0.23.3 route-map PREPEND out
route-map PREPEND permit 10
set as-path prepend 200 200 200
```

After applying the route-map, R3 will see the AS_PATH as 200 200 200 100, making it longer and thus less preferred compared to other available paths.

Conclusion:

This practical demonstrates the significance of the AS_PATH attribute in BGP for both loop prevention and path selection. By manipulating the AS_PATH using techniques like AS Path Prepending, network administrators can influence routing decisions to achieve desired traffic engineering outcomes. The GNS3 simulation effectively illustrates how BGP routers use the AS_PATH attribute to determine the best path to a destination network.

For video demonstration of the above practical scan the following QR-code or type the link address

<https://youtu.be/28yj646Wwro?si=WRZ4ePgWTNJQ-zWP>



Configuring IBGP and EBGP Sessions, Local Preference, and MED

Aim: To configure IBGP, EBGP and set the local preference and MED

Theory:

Border Gateway Protocol (BGP) Overview

BGP (Border Gateway Protocol) is a routing protocol used to exchange routing information between different networks on the internet. It is classified into two types:

- **External BGP (EBGP)** – Used between different Autonomous Systems (AS).
- **Internal BGP (IBGP)** – Used within the same Autonomous System.

External BGP (EBGP)

- EBGP is used to exchange routes between different Autonomous Systems (ASes).
- It is commonly used by Internet Service Providers (ISPs) and large enterprises to communicate with external networks.
- The default Time-To-Live (TTL) value is 1, meaning that EBGP peers must be directly connected unless explicitly configured otherwise.
- AS-path attribute is used in EBGP to prevent routing loops.
- It prefers shorter AS paths when selecting the best route.
- **Example Scenario:**
If AS100 wants to exchange routes with AS200, they establish an EBGP connection between their routers.

Internal BGP (IBGP)

- IBGP is used for routing within the same Autonomous System.
- It ensures that all routers in an AS have a consistent view of external routes learned via EBGP.
- Unlike EBGP, IBGP does not modify the AS-path attribute.
- IBGP requires a full mesh of connections (or Route Reflectors/Confederations to reduce overhead).
- Next-hop attribute must be reachable within the AS for proper routing.
- **Example Scenario:**
If AS100 has multiple routers, they must use IBGP to share routes learned from EBGP peers.

Key Differences Between EBGP and IBGP

Feature	EBGP	IBGP
Used for	Between different ASes	Within the same AS
AS-Path Modification	Yes	No
Next-Hop Change	Yes	No (next-hop must be reachable)
Default TTL	1	255
Full Mesh Required?	No	Yes (or use Route Reflectors)

MED and Local Preference in BGP

BGP (Border Gateway Protocol) uses several attributes to influence routing decisions. Two important attributes that help in path selection are **MED (Multi-Exit Discriminator)** and **Local Preference**.

Multi-Exit Discriminator (MED)

Purpose: MED is used to influence the incoming traffic from an external AS by suggesting the preferred entry point into an AS when multiple links exist.

Characteristics:

- It is an **optional, non-transitive** attribute.
- A **lower MED value** is preferred.
- MED is shared only with **directly connected external neighbors** (not propagated beyond the next AS).
- It is commonly used between ISPs or between enterprise networks and ISPs.
- **Example Scenario:**
If AS100 has two links to AS200, it can set a lower MED on one link to tell AS200 to prefer that path for incoming traffic.

Local Preference

Purpose: Local Preference is used within an AS to influence the outgoing traffic by selecting the preferred exit point when multiple paths to the same destination exist.

Characteristics:

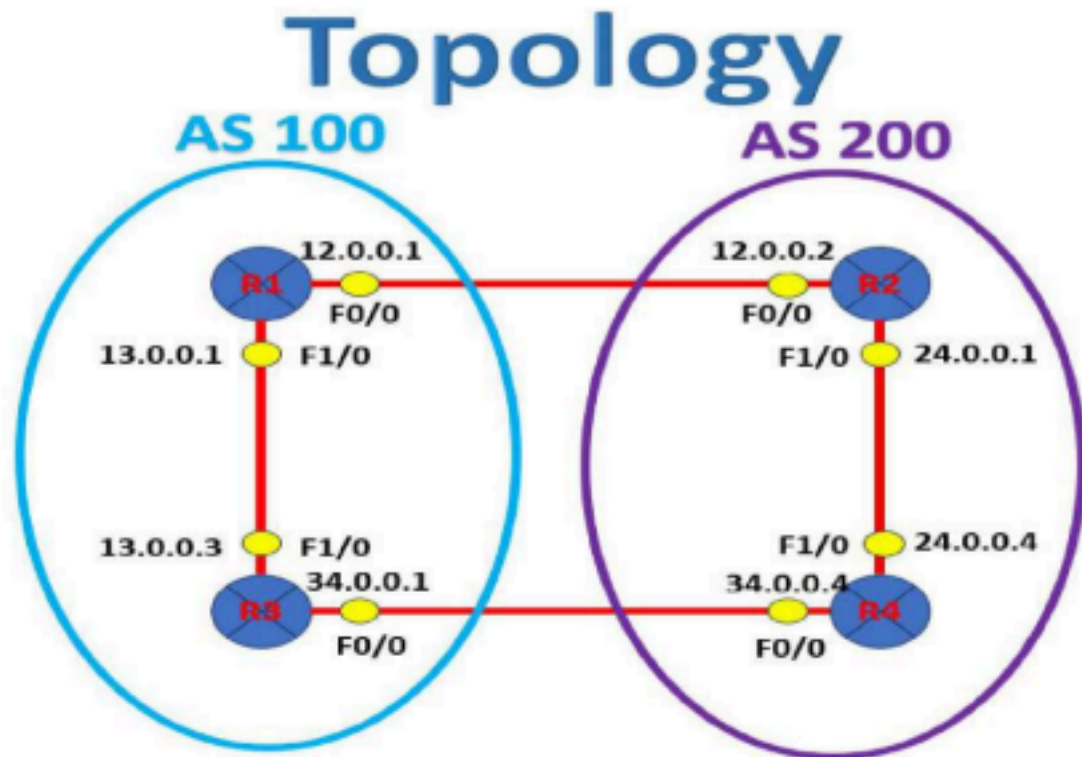
- It is a **well-known, discretionary** attribute.
- A **higher Local Preference value** is preferred.
- It is propagated within the AS to all IBGP peers.
- Used mainly by ISPs and large networks to control outbound traffic flow.
- **Example Scenario:**
If AS100 has two exit points (R1 and R2) to AS200, setting a higher Local Preference on R1 will make all routers in AS100 prefer R1 for outgoing traffic.

Key Differences Between MED and Local Preference

Feature	MED (Multi-Exit Discriminator)	Local Preference
Function	Controls incoming traffic from another AS	Controls outgoing traffic within an AS
Preference Rule	Lower value preferred	Higher value preferred
Scope	Shared with EBGp peers but not propagated further	Propagated to all IBGP peers
Attribute Type	Optional, non-transitive	Well-known, discretionary
Used By	External ASes to choose entry points	Internal AS to choose exit points

Both attributes play a crucial role in BGP traffic engineering by influencing how traffic enters and exits an autonomous system.

We use the following topology



We do the configuration using the following steps

Step 1: Configure the IP addresses on all the Routers

Router 1

```

R1#
R1#configure terminal
R1(config)#
R1(config)#interface fastEthernet 0/0
R1(config-if)#
R1(config-if)#ip address 12.0.0.1 255.255.255.0
R1(config-if)#
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#exit
R1(config)#
R1(config)#interface fastEthernet 1/0
R1(config-if)#ip address 13.0.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
  
```

```
R1(config-if)#exit
R1(config)
```

Router 2

```
R2#
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#interface fastEthernet 0/0
R2(config-if)#
R2(config-if)#ip address 12.0.0.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface fastEthernet 1/0
R2(config-if)#
R2(config-if)#ip address 24.0.0.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
```

Router 3

```
R3#
R3#configure terminal
R3(config)#i
R3(config)#interface fastEthernet 0/0
R3(config-if)#
R3(config-if)#ip address 34.0.0.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#
R3(config-if)#exit
R3(config)#
R3(config)#interface fastEthernet 1/0
R3(config-if)#
R3(config-if)#ip address 13.0.0.3 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
```

Router4

```
R4#
R4#configure terminal
R4(config)#
```

```

R4(config)#interface fastEthernet 0/0
R4(config-if)#
R4(config-if)#ip address 34.0.0.4 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
R4(config)#interface fastEthernet 1/0
R4(config-if)#
R4(config-if)#ip address 24.0.0.4 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#

```

Step 2: Set IBGP and EBGp on each router

Router 1

```

R1(config)#router bgp 100
R1(config-router)#
R1(config-router)#neighbor 13.0.0.3 remote-as 100
R1(config-router)#neighbor 13.0.0.3 update-source fastEthernet 1/0
R1(config-router)# network 12.0.0.0 mask 255.255.255.0
R1(config-router)#neighbor 12.0.0.2 remote-as 200
R1(config-router)#network 13.0.0.0 mask 255.255.255.0
R1(config-router)#
R1(config-router)#exit
R1(config)#exit
R1#

```

Router 2

```

R2(config)#router bgp 200
R2(config-router)#
R2(config-router)#neighbor 24.0.0.4 remote-as 200
R2(config-router)#neighbor 24.0.0.4 update-source fastEthernet 1/0
R2(config-router)#network 12.0.0.0 mask 255.255.255.0
R2(config-router)#neighbor 12.0.0.1 remote-as 100
R2(config-router)#network 24.0.0.0 mask 255.255.255.0
R2(config-router)#exit
R2(config)#

```

Router 3

```

R3(config)#router bgp 100
R3(config-router)#
R3(config-router)#neighbor 13.0.0.1 remote-as 100
R3(config-router)#neighbor 13.0.0.1 update-source fastEthernet 1/0

```

```
R3(config-router)#network 34.0.0.0 mask 255.255.255.0
R3(config-router)#neighbor 34.0.0.4 remote-as 200
R3(config-router)#network 13.0.0.0 mask 255.255.255.0
R3(config-router)#
```

Router 4

```
R4(config)#
R4(config)#router bgp 200
R4(config-router)#
R4(config-router)#neighbor 24.0.0.1 remote-as 200
R4(config-router)#neighbor 24.0.0.1 update-source fastEthernet 1/0
R4(config-router)#network 34.0.0.0 mask 255.255.255.0
R4(config-router)#neighbor 34.0.0.3 remote-as 100
R4(config-router)#network 24.0.0.0 mask 255.255.255.0
R4(config-router)#exit
R4(config)#
```

Step 4: Verify the BGP protocol by pinging from Router1 to all interfaces

R1#ping 12.0.0.2

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/29/36 ms
```

R1#ping 13.0.0.3

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 13.0.0.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/26/36 ms
```

R1#ping 24.0.0.4

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 24.0.0.4, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/60/64 ms
```

R1#ping 34.0.0.4

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 34.0.0.4, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/56/60 ms
```


Step 5: Configure Local Preference (Influencing Outbound Traffic from AS 100)

```

R1#
R1#configure terminal
R1(config)#
R1(config)#route-map smile_ip permit 10
R1(config-route-map)#
R1(config-route-map)#set local-preference 200
R1(config-route-map)#
R1(config-route-map)#exit
R1(config)#router bgp 100
R1(config-router)#
R1(config-router)#neighbor 13.0.0.3 route-map smile_ip in
R1(config-router)#exit
R1(config)#exit

```

Step 6: Configure MED (Influencing Inbound Traffic to AS 100)

```

R2(config)#
R2(config)#route-map set_med permit 10
R2(config-route-map)#set metric 50
R2(config-route-map)#exit
R2(config)#
R2(config)#router bgp 200
R2(config-router)#
R2(config-router)#neighbor 12.0.0.1 route-map set_med out
R2(config-router)#exit
R2(config)#

```

Step 7: Verification

```

R1#
R1#show ip bgp summary

```

```

BGP router identifier 13.0.0.1, local AS number 100
BGP table version is 7, main routing table version 7
4 network entries using 576 bytes of memory
6 path entries using 480 bytes of memory
3/3 BGP path/bestpath attribute entries using 408 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1488 total bytes of memory
BGP activity 4/0 prefixes, 6/0 paths, scan interval 60 secs

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
12.0.0.2	4	200	19	15	7	0	0	00:08:42	3
13.0.0.3	4	100	12	15	7	0	0	00:07:41	2

R1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
 C 12.0.0.0/24 is directly connected, FastEthernet0/0
 L 12.0.0.1/32 is directly connected, FastEthernet0/0
 13.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
 C 13.0.0.0/24 is directly connected, FastEthernet1/0
 L 13.0.0.1/32 is directly connected, FastEthernet1/0
 24.0.0.0/24 is subnetted, 1 subnets
 B 24.0.0.0 [20/50] via 12.0.0.2, 00:00:24
 34.0.0.0/24 is subnetted, 1 subnets
 B 34.0.0.0 [200/0] via 13.0.0.3, 00:07:31

R1#show ip bgp

BGP table version is 7, local router ID is 13.0.0.1
 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
r> 12.0.0.0/24	12.0.0.2	50	0	200	i
* i 13.0.0.0/24	13.0.0.3	0	100	0	i
*> 0.0.0.0		0	32768		i
*> 24.0.0.0/24	12.0.0.2	50	0	200	i
* 34.0.0.0/24	12.0.0.2	50	0	200	i
*>i 13.0.0.3		0	100	0	i



























































